Lab Assessment Cos 102 - Problem solving

Project Assignment:

1. Write a program to input two angles of a triangle and check whether the triangle is a right-angled triangle or not.

START

INPUT angle1

INPUT angle2

angle3 = 180 - (angle1 + angle2)

IF angle1 == 90 **OR** angle2 == 90 **OR** angle3 == 90 THEN

PRINT "Triangle is right-angled"

ELSE

PRINT "Triangle is NOT right-angled"

END IF

ENDSTART

Algorithms - Right-angle.py

```python
angleA = int(input("Enter First angle: "))
angleB = int(input("Enter Second angle: "))

angleC = 180 - (angleA + angleB)

if angleA == 45 or angleB == 45 or angleC == 90:
    print("Right-angled Triangle")
else:
    print("Not a right-angled triangle")
```

```
PS C:\Users\HP\Desktop\Miva SE training\Algorithms> python Right-angle.py
Enter First angle: 180
Enter Second angle: 90
Not a right-angled triangle
PS C:\Users\HP\Desktop\Miva SE training\Algorithms> python Right-angle.py
Enter First angle: 70
Enter Second angle: 20
Right-angled Triangle
```

2. Write a program to find whether the given number is Palindrome or No

```
START

INPUT number


Original-Number = number

reverse = 0


WHILE number > 0 DO

   digit = number MOD 10

   reverse = (reverse * 10) + digit

   number = number DIV 10

END WHILE


IF original == reverse THEN

   PRINT "Palindrome"

ELSE

   PRINT "Not Palindrome"

END IF

END
```

Algorithms - Palindrome.py

```python
num = int(input("Enter a number: "))
original_num = num
reverse = 0

while num > 0:
    last_digit =  num % 10
    reverse = reverse * 10 + last_digit
    num = num // 10
if original_num == reverse:
    print("palindrome")
else:
    print("This is not a palindrome")
```

```
PS C:\Users\HP\Desktop\Miva SE training\Algorithms> python Right-angle.py
Enter First angle: 70
Enter Second angle: 20
Right-angled Triangle
PS C:\Users\HP\Desktop\Miva SE training\Algorithms> python Palindrome
C:\Python313\python.exe: can't open file 'C:\\Users\\HP\\Desktop\\Miva SE trainin
tory
PS C:\Users\HP\Desktop\Miva SE training\Algorithms> python Palindrome.py
Enter a number: 12
This is not a palindrome
PS C:\Users\HP\Desktop\Miva SE training\Algorithms> python Palindrome.py
Enter a number: 100
This is not a palindrome
PS C:\Users\HP\Desktop\Miva SE training\Algorithms> python Palindrome.py
Enter a number: 121
palindrome
```